Sprint - 7 Report

Project Name: Personalized Study Planner

Team Name: Group-7

Scrum Master: Aashish

Sprint Duration: 2 weeks

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1. Functional Testing:

Implementation Strategy:
The team conducted functional testing by combining automated JUnit tests (already written in the Spring Boot backend) with structured manual testing of the React frontend. After the code cleanup performed in Sprint 6, a final round of unit tests was executed to detect any regressions introduced during refactoring. Test results were tracked in a bug table, and identified issues were assigned, fixed, and re-tested within the sprint.

Jenkins was used to build the project, run the test suite, and generate the SonarQube static code analysis report. Docker was used to verify the full application stack (backend, frontend, PostgreSQL) could be deployed successfully as containers.

Deliverables:

- **Test Plan**: covering objectives, resources, test environment, and all functional and non-functional test tasks
- **JUnit Test Execution Report**: final run of all unit tests after Sprint 6 code cleanup, confirming all tests pass
- **Bug Tracking Table**: all identified defects logged with severity, status, and fix description
- **Trello Update**: user stories reviewed and moved to appropriate columns reflecting Sprint 7 completion status
- **Jenkins Build Report**: successful project build with SonarQube static analysis showing all grades A or B, and successful Docker deployment

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2. Non-functional Testing

1. Heuristic evaluation

Implementation Strategy: Team members evaluated the application against Nielsen's 10 Usability Heuristics, walking through all major user flows. Findings were rated by severity (0 - 4) and documented with recommendations. Average severity was 1.5/4, no critical issues found. Key findings included missing save confirmations, inconsistent button labels, and lack of delete confirmation dialogs.

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2. User Acceptance Testing

Implementation Strategy: UAT was conducted using the Sprint 6 UAT template. Team members tested six core user scenarios, registration/login, task management, session scheduling, calendar view, language switching, and profile management. All scenarios passed. One localization bug (language reset on refresh) was identified and fixed during the sprint.

3.Static Code Analysis

Implementation Strategy: SonarQube was integrated into the Jenkins pipeline. After each build, the scanner analyzed the Spring Boot backend for bugs, vulnerabilities, code smells, and duplications. All quality gate metrics were resolved to grade A or B before sprint completion. *(Screenshots attached from Jenkins dashboard.)*

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4. Security Testing report:

Implementation Strategy: Manual security testing was performed based on OWASP Top 10 fundamentals, covering authentication, authorization, input validation, and API configuration. Key findings included an overly permissive CORS configuration (wildcard * origin) which was fixed, and a missing rate limit on login, noted as a recommendation for production. All other areas (JWT validation, SQL injection, XSS, password hashing) passed without issues.

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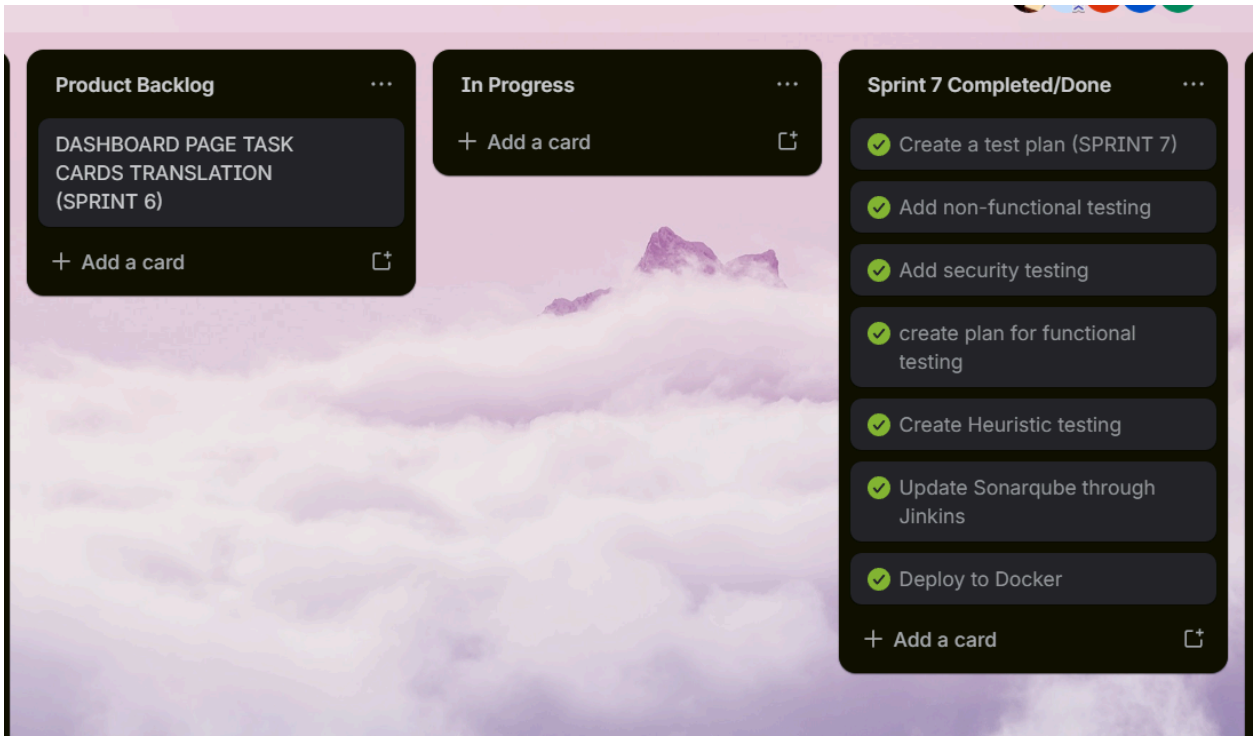
3. Members Contributions :

Name	Assigned Task	Hours taken	In-class assignment
Aashish	Scrum Master, Test Plan, Sprint coordination, Trello updates	15	Submitted
Jari	UAT execution, bug tracking table	15	Submitted
Megha	Security testing, technical changes documentation	15	Submitted
Mhin	Heuristic Evaluation, documentation	15	Submitted
Mustafa	JUnit testing, bug fixing, Jenkins/SonarQube pipeline	15	Submitted

Trello ScreenShort: